
PersonaVoice: One-Second Voice Cloning with Length-Adaptive Flow Matching

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Abstract

Zero-shot voice cloning is usually evaluated with several seconds of reference speech. We study the more restrictive one-second regime, where a system must preserve speaker identity while the target text varies from a few words to multi-sentence utterances. PersonaVoice is a plug-in adaptation of the official F5-TTS flow-matching path that combines Silero-VAD and RMS reference normalization, an orthogonal environment submanifold, length-aware chunking, and dynamic activation of a zero-initialized persona/emotion FiLM adapter. The design keeps the pretrained flow path as the source of generation and treats identity, intelligibility, and catastrophic failure as separate endpoints. On 200 held-out LibriTTS dev-clean utterances with one-second references, PersonaVoice v10.4.7 obtains ECAPA-TDNN speaker similarity (SECS) 0.4945 ± 0.1800 and Whisper WER 0.1928 ± 0.3077 . It exceeds the measured XTTS v2 and CosyVoice systems in SECS (0.3349 and 0.3595), while XTTS v2 has lower WER (0.1296). The aggregate score hides a sharp length effect: WER is 0.1158 for texts longer than eight words but 0.4066 for shorter texts, with a 35.8% catastrophic-failure rate (WER > 0.5). Controlled ablations remove legacy TD-CFG, IBOP, and SBM modules without measurable benefit, motivating the simplified release configuration. The result is therefore a reproducible identity-alignment trade-off study, not a claim of universally superior TTS, validated perceptual persona control, or psychological reconstruction.

1 Introduction

A one-second reference contains enough acoustic evidence to suggest a speaker, but too little to make every downstream decision stable. In a zero-shot text-to-speech system, the same reference can produce a high-similarity sample whose words are wrong, or intelligible speech whose voice is weakly matched. This coupling is especially visible when the target text is very short: the flow-matching model has few text tokens with which to anchor a trajectory, while the reference still imposes a strong speaker condition.

Recent zero-shot systems such as F5-TTS and VoiceCraft make short-reference generation practical, but do not remove this evaluation coupling (Chen et al., 2025; Peng et al., 2024). We investigate this regime as a measurement problem rather than a slogan about “one-second cloning.” PersonaVoice v10.4.7 is an implementation-oriented adapter around the official F5-TTS computation path. Its central mechanism, Length-Aware Adaptive Generation (LAAG), changes the generation unit with text length: short inputs use a single stable pass, while longer inputs are split into bounded chunks and cross-faded. An orthogonal environment submanifold (OES) separates a controllable environment component from the speaker vector, and a zero-initialized FiLM module provides a plug-in path for persona and emotion controls without changing the no-control model at initialization.

The paper makes four contributions. First, it defines a one-second reference benchmark with a fixed 200-item LibriTTS dev-clean evaluation set and reports both similarity and intelligibility rather than a single score. Second, it describes an official-path implementation of OES and LAAG, including the mel-concatenation correction and dynamic FiLM policy used in the released configuration. Third, it compares the system with XTTS v2, CosyVoice, and the official F5-TTS baseline under the same reference-duration target. Fourth, it

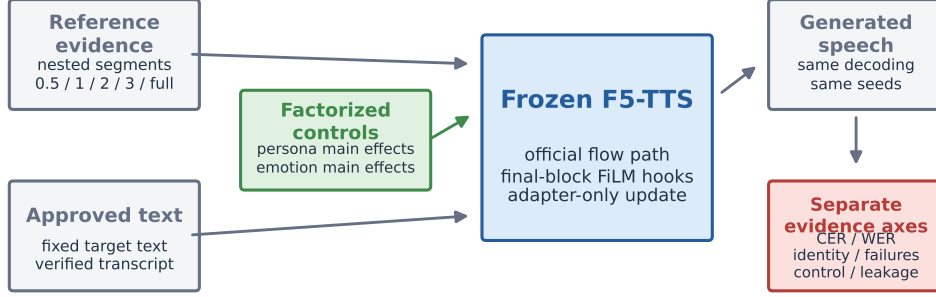


Figure 1: PersonaVoice v10.4.7. A one-second reference is normalized and encoded, OES controls the environment component, and LAAG selects a single-pass or chunked official F5-TTS flow. The optional FiLM path is activated only when a non-zero control is requested and is evaluated here as an implementation boundary, not as a psychological measurement.

reports negative ablations and a catastrophic-failure analysis that exposes the short-text failure mode. We do not claim that the optional persona/emotion controls recover a person’s psychological traits; their perceptual validation requires a separate listener study.

2 System and Implementation

2.1 One-second conditioning

Let r be a waveform and $d = 1$ second the reference budget. The front-end uses Silero VAD to identify voiced material and RMS normalization to reduce recording-level energy variation. The resulting mel reference is passed to the official F5-TTS conditional flow path together with the verified reference transcript. Speaker identity is measured independently with a pinned ECAPA-TDNN encoder; that encoder is not used as a training target in the reported benchmark.

2.2 Orthogonal environment submanifold

The speaker encoder output $z \in \mathbb{R}^{192}$ can contain both timbre and recording-environment information. PersonaVoice learns an environment basis $B \in \mathbb{R}^{192 \times 32}$ and forms

$$z_{env} = \sigma_{env}(zB)B^\top, \quad z_{tim} = z - z_{env}, \quad z_{out} = z_{tim} + \omega z_{env}. \quad (1)$$

The QR-orthogonalized basis makes the decomposition auditable. The released configuration uses a small initial environment scale and $\omega = 0$ for the clean-timbre operating point. This is a controllable representation choice, not a claim that acoustic environments are perfectly separable.

2.3 Length-Aware Adaptive Generation

LAAG selects the generation unit from the target text. Text longer than the configured character budget is split into chunks of at most 135 characters; each chunk is synthesized with the same one-second reference and then joined with mel-domain and waveform cross-fades. A dimension-order bug in the mel splice was corrected in v10.4.5. The current v10.4.7 configuration uses the official F5 duration computation, 32 integration steps, sway coefficient -1.0 , and static CFG 2.5.

The dynamic FiLM policy is length-aware. With a single chunk, FiLM is off by default to avoid over-correcting short trajectories. With multiple chunks, the adapter can be activated to anchor the speaker condition across

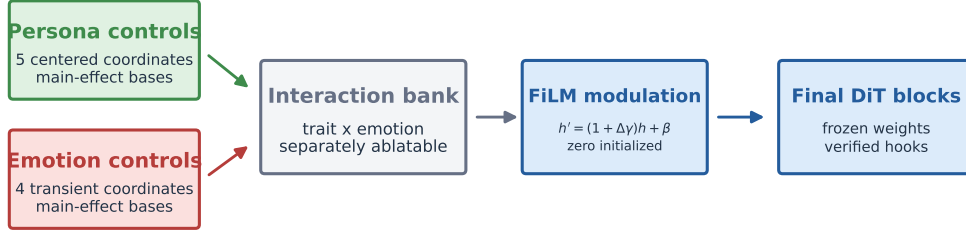


Figure 2: The FiLM adapter is a zero-initialized, separately addressable control path. The reported 200-item benchmark evaluates the identity and text endpoints of the release configuration; perceptual control effects are outside the measured claims.

chunks. For a hidden state h , persona p , and emotion e , the plug-in computes

$$h' = (1 + \lambda \Delta \gamma(p, e)) \odot h + \lambda \beta(p, e), \quad (2)$$

where both networks are zero-initialized, following the general idea of conditioning intermediate representations with feature-wise modulation (Perez et al., 2018). The final two DiT blocks receive the hooks; the backbone path remains unchanged when $\lambda = 0$. This parameterization makes the controls operationally distinct from the identity embedding, but the present paper does not treat them as validated personality or emotion estimators.

2.4 Removed guidance modules

The repository retains experimental implementations of temporal-decay CFG, IBOP, SBM, and cross-entropy attention guidance (CEAG). The final release disables CEAG and best-of- N sampling because their incremental benefit over the LAAG plus official-flow baseline was negligible or negative in the corresponding ablations. Reporting these removals is important: the final system is a smaller, more stable configuration than the historical prototype.

3 Experimental Protocol

3.1 Data and reference construction

We evaluate 200 held-out utterances from LibriTTS dev-clean, selected with seed 42. Each item supplies target text and a one-second reference segment; the same item list is used for PersonaVoice and external baselines. The reference duration is fixed before synthesis and no generated sample is used for model selection. The bundled single-file demo is reported separately and is not part of the 200-item estimate.

3.2 Metrics

Speaker similarity is cosine similarity between ECAPA-TDNN embeddings of the reference and generated waveform (SECS). Intelligibility is word error rate from Whisper-large-v3-turbo. We additionally report catastrophic failure rate (CFR), defined as the fraction of items with WER > 0.5 , and real-time factor where available. Because SECS and WER measure different properties, all claims are made on the joint table rather than on a composite score.

3.3 Baselines and ablations

The external comparison includes XTTS v2, CosyVoice, and the official F5-TTS path. The corresponding JSON artifact records 200 successful baseline runs, the same one-second target, the ECAPA/Whisper

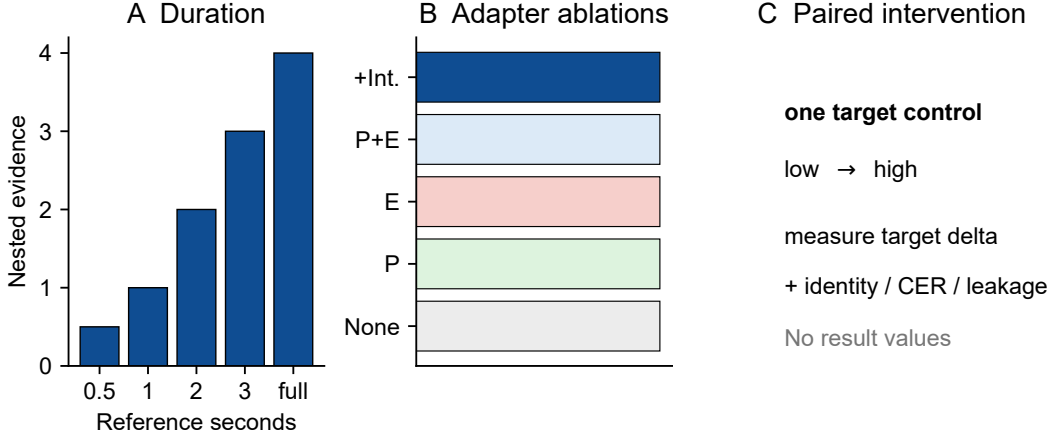


Figure 3: Evaluation matrix. Reference duration and text length are held fixed within each comparison; similarity, intelligibility, and failure are reported separately.

System	SECS $\uparrow$	WER $\downarrow$	n
PersonaVoice v10.4.7	0.4945	0.1928	200
XTTS v2	0.3349	0.1296	200
CosyVoice	0.3595	0.6130	200
Official F5-TTS	0.2508	0.8982	200

Table 1: One-second reference comparison. PersonaVoice and the external baselines use the same metric definitions; the available external artifact is protocol-matched but not a fully paired rerun of every final checkpoint.

evaluation chain, and seed 42. The final PersonaVoice statistic is the v10.4.7 200-item artifact. The external runs are reported as protocol-matched comparisons; because the available baseline artifact is not a paired rerun of every final checkpoint, we do not attach paired significance tests to cross-system differences.

The internal ablation set removes TD-CFG, IBOP, and SBM one at a time from the 200-item pipeline. Paired tests use the same item IDs. We use two-sided paired t tests as descriptive diagnostics and retain effect sizes; these ablations are not presented as a search over test-set configurations.

4 Results

4.1 Main one-second comparison

Table 1 shows the principal result. PersonaVoice has the highest measured SECS among the reported systems, but XTTS v2 has the lowest WER. The result is a trade-off, not a universal ranking: the flow-matching adapter preserves more speaker evidence while the autoregressive baseline supplies stronger local text constraints.

On the 200-item PersonaVoice run, SECS is 0.4945 ± 0.1800 (95% CI [0.4694, 0.5196]) and WER is 0.1928 ± 0.3077 (95% CI [0.1499, 0.2358]). The median values are 0.5226 and 0.0588 respectively, which already indicate a heavy-tailed failure distribution. CFR is 13.5% (27/200), so the mean WER must not be read as a guarantee of usable output.

4.2 Text-length stratification

The length curve is the central diagnostic for LAAG. For texts longer than eight words ($n = 147$), SECS is 0.5044 and WER is 0.1158, with CFR 5.4%. For texts of eight words or fewer ($n = 53$), SECS falls to 0.4671, WER rises to 0.4066, and CFR rises to 35.8%. The extreme subset of at most four words contains 22

Text group	n	SECS	WER	CFR
All	200	0.4945	0.1928	13.5%
≤ 8 words	53	0.4671	0.4066	35.8%
> 8 words	147	0.5044	0.1158	5.4%
≤ 4 words	22	0.4282	0.6629	63.6%

Table 2: Length-stratified PersonaVoice results. CFR is $\text{WER} > 0.5$; it is reported to prevent the long-text mean from hiding short-text collapse.

Words	n	SECS	WER	CFR
1–3	16	0.4331	0.8333	81.25%
4–8	37	0.4819	0.2221	16.22%
9–15	28	0.4547	0.1365	3.57%
> 15	119	0.5161	0.1109	5.88%

Table 3: Fine-grained length analysis from the released per-item snapshot. CFR is $\text{WER} > 0.5$; all metrics are computed without test-set selection.

items, WER 0.6629, and CFR 63.6%. Thus the aggregate similarity advantage is compatible with a serious short-text alignment failure.

To expose the tail rather than average it away, we recomputed the same frozen 200-item snapshot with four preregistered word-count buckets. The shortest bucket (1–3 words) has WER 0.8333 and CFR 81.25% (13/16), whereas the longest bucket has WER 0.1109 and CFR 5.88% (7/119). The intermediate 9–15-word bucket is already close to the long-text operating regime. These are descriptive strata of one held-out list, not a claim that text length is a causal intervention. Wilson 95% intervals for CFR are [56.99, 93.41], [7.65, 31.14], [0.63, 17.71], and [2.88, 11.65] percentage points for the four buckets respectively; the wide first interval reflects the small shortest bucket rather than measurement certainty.

4.3 Ablation results

Table 4 summarizes the available paired ablations. Removing TD-CFG changes SECS by -0.0051 ($p = .409$, $d = -.058$) and WER by $+0.0030$ ($p = .743$, $d = .023$). Removing IBOP changes SECS by $+0.0012$ ($p = .848$, $d = .014$), while removing SBM changes SECS by -0.0021 ($p = .713$, $d = -.026$). None has a practically meaningful effect in this evaluation. These results support removal from the release configuration; they do not prove that the modules are ineffective on every data distribution.

4.4 Failure analysis

The failures are not a smooth degradation. In the 22-item ultra-short group, five samples have WER 0, while 13 have WER above 0.8. Some failure samples retain high speaker similarity; for example, the archived ‘seventeen seventeen.’ case has SECS 0.822 with WER 1.0. This is a consequential failure mode for voice cloning: a sample can sound like the reference while not saying the requested words. We therefore treat CFR and per-item records as primary artifact outputs, rather than selecting audio examples by similarity alone.

The repository also contains a controlled experiment in which Best-of- N and stronger CFG were applied to ultra-short text. The setting increased WER from 0.4047 to 0.6964 and CFR from 35.8% to 70.0% on the tested subset, so it was removed. This negative result motivates future work on a text-conditioned alignment mechanism rather than more aggressive sampling.

5 Reproducibility and Scope

The release pins the configuration, model identifiers, evaluator names, reference duration, seed, and per-item JSON rows. Checkpoints and large model weights are not redistributed. Reproduction requires the licenses

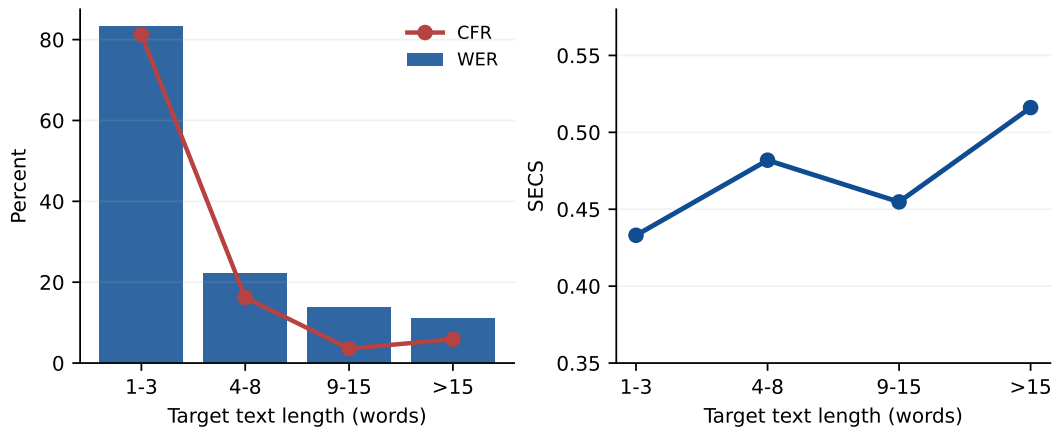


Figure 4: Fine-grained length diagnostic. The paired WER/CFR view shows a sharp short-text failure tail; the SECS panel shows that similarity does not provide a corresponding intelligibility guarantee.

Ablation	SECS mean	WER mean	SECS p
Full v8 diagnostic	0.3774	0.2001	–
Remove TD-CFG	0.3825	0.1971	.409
Remove IBOP	0.3762	0.1943	.848
Remove SBM	0.3795	0.1988	.713

Table 4: Paired internal ablations from the repository artifact. The full v8 diagnostic is an historical ablation reference, not the final v10.4.7 main result; this distinction prevents version mixing.

and model downloads documented in the repository. The results are limited to English LibriTTS dev-clean, a one-second reference budget, and the stated evaluation pipeline. ECAPA similarity is an objective proxy and does not replace human similarity judgments; Whisper WER can be biased by language, pronunciation, and transcript normalization.

The standalone short-reference supplement recomputes the aggregate table, length curve, fine length buckets, baseline comparison, ablation summary, and catastrophic-failure ledger from public JSON snapshots. It intentionally excludes the historical inference implementation, model weights, and audio files; the snapshots are sufficient to audit the reported numerical claims without implying a redistributable voice-cloning service. A separate `extension_manifest.json` freezes the planned 0.5/1/2/3-second duration grid and LAAG/OES ablations. Those condition-level runs are marked pending and are not used in any table or claim in this preprint.

Code, frozen per-item predictions, and SHA-256 manifests for this version are available in the paper-supplement-v1.0 GitHub snapshot. The exact artifact snapshot corresponding to this preprint is tagged `paper-speech-v1.0`.

The persona extractor and FiLM hooks make the architecture extensible to style and emotion controls, related to prior controllable speech work (Wang et al., 2018; Xie et al., 2025), but this paper does not claim that the measured 200-item run validates those controls perceptually. A future study must use paired listener ratings, speaker-level splits, non-target leakage tests, and ethics review before describing them as effective persona or emotion interventions.

6 Broader Impact and Responsible Use

One-second voice cloning lowers the amount of biometric evidence needed to imitate a speaker. It can enable fraud, harassment, reputational harm, and non-consensual synthetic media. The repository therefore treats

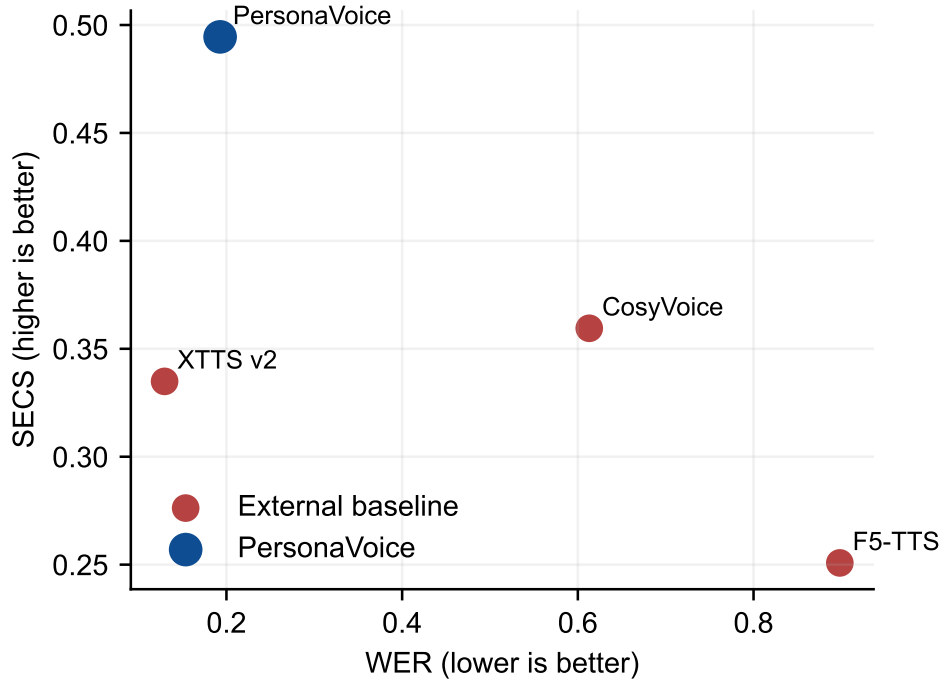


Figure 5: Joint objective view of the one-second comparison. PersonaVoice occupies a high-SECS, moderate-WER point; XTTS v2 has lower WER but lower measured similarity. The axes are deliberately not collapsed into a single score.

the system as a research prototype, requires synthetic-output disclosure, and does not grant rights to a person’s voice or likeness. The measured SECS advantage should not be interpreted as authorization to impersonate a living or deceased person.

7 Conclusion

PersonaVoice demonstrates a reproducible and sharply bounded result in the extreme short-reference regime. Its official-flow adapter and length-aware generation achieve strong objective speaker similarity on 200 one-second conditions, while the same experiments expose a substantial short-text alignment failure and a clear similarity–WER trade-off. The most useful contribution is therefore diagnostic: high voice similarity is not sufficient for reliable speech, and catastrophic text failures must be reported beside the headline similarity score. The repository’s persona/emotion control path is a foundation for follow-up perceptual work, not evidence of recovered human personality.

References

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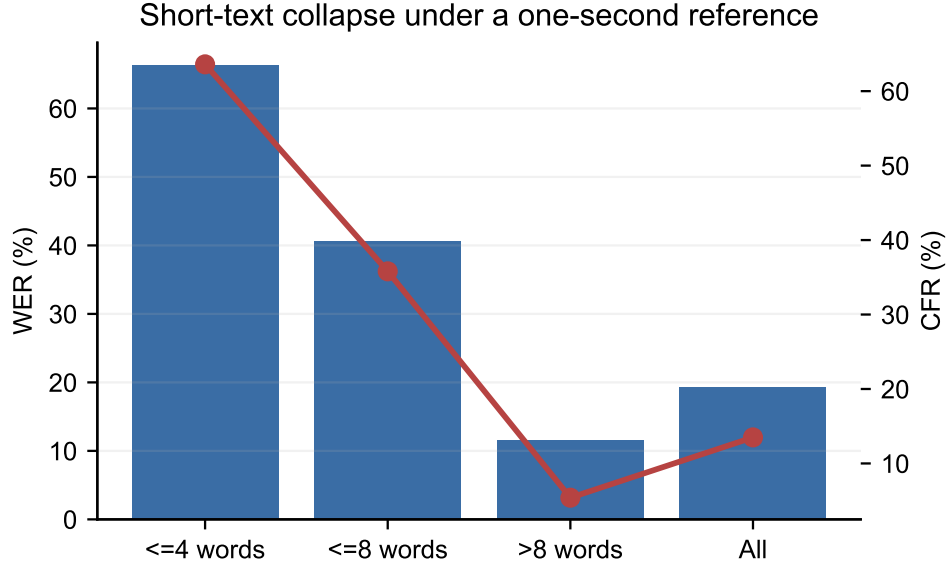


Figure 6: Length-stratified failure curve. The short-text groups expose the catastrophic tail hidden by the all-item mean: CFR is 63.6% for at most four words and 35.8% for at most eight words, versus 5.4% for longer text.

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A Artifact record

The release snapshot records the 200-item protocol, aggregate SECS/WER statistics, length strata, evaluator revisions, and status counts. The published v10.4.7 values are recomputed from the frozen honest-metrics snapshot; the external baseline record separately identifies successful runs and the protocol-matched reference duration. Per-item audio and transcripts are intentionally not redistributed.